

1 Solution — GIAM 4.3

Problem 4

1.1 Problem

For each positive integer n , define an interval I_n by

$$I_n = \left[-n, \frac{1}{n}\right).$$

Find the union and the intersection of all the intervals in this infinite family.

1.2 Answer

$$\begin{aligned} \bigcup_{n \in \mathbb{N}} I_n &= (-\infty, 1) \\ \bigcap_{n \in \mathbb{N}} I_n &= [-1, 0] \end{aligned}$$

Both were checked against a spread of test points — including -5000 , $\frac{1}{10000}$, $\frac{999}{1000}$ and the four endpoints — with the search bound chosen large enough for each point separately.

1.3 The Union

$$x \in \bigcup_n I_n \iff \exists n \in \mathbb{N} : -n \leq x \text{ and } x < \frac{1}{n}.$$

Containment $\subseteq$. If $x \in I_n$ for some n then $x < \frac{1}{n} \leq 1$, so $x \in (-\infty, 1)$.

Containment $\supseteq$. Let $x < 1$. If $x \geq -1$, then $x \in I_1 = [-1, 1)$ and we are done. If $x < -1$, choose any integer $n \geq -x$; then $-n \leq x$, and since $x < 0 < \frac{1}{n}$ the right condition holds too, so $x \in I_n$.

Therefore the union is $(-\infty, 1)$. The right endpoint 1 is never attained, since $\frac{1}{n} \leq 1$ for every n and each interval excludes its right endpoint.

1.4 The Intersection

Here it is cleanest to split each interval into two one-sided conditions, $I_n = [-n, \infty) \cap (-\infty, \frac{1}{n})$, because intersection distributes over intersection:

$$\bigcap_n I_n = \left(\bigcap_n [-n, \infty) \right) \cap \left(\bigcap_n (-\infty, \frac{1}{n}) \right).$$

Left factor. $-n \leq x$ for every $n \geq 1$ is tightest at $n = 1$, so the condition is just $x \geq -1$ and the left factor is $[-1, \infty)$.

Right factor. $x < \frac{1}{n}$ for every n holds exactly when $x \leq 0$. Indeed if $x \leq 0$ then $x \leq 0 < \frac{1}{n}$ for all n ; and if $x > 0$, choosing $n > \frac{1}{x}$ gives $\frac{1}{n} < x$, so $x \notin I_n$. The right factor is $(-\infty, 0]$.

Intersecting, $\bigcap_n I_n = [-1, \infty) \cap (-\infty, 0] = [-1, 0]$.

1.5 The Picture

The five smallest intervals are drawn below on a common axis, with the two answers underneath them. Reading down the left endpoints and then down the right endpoints makes both computations visible at once.

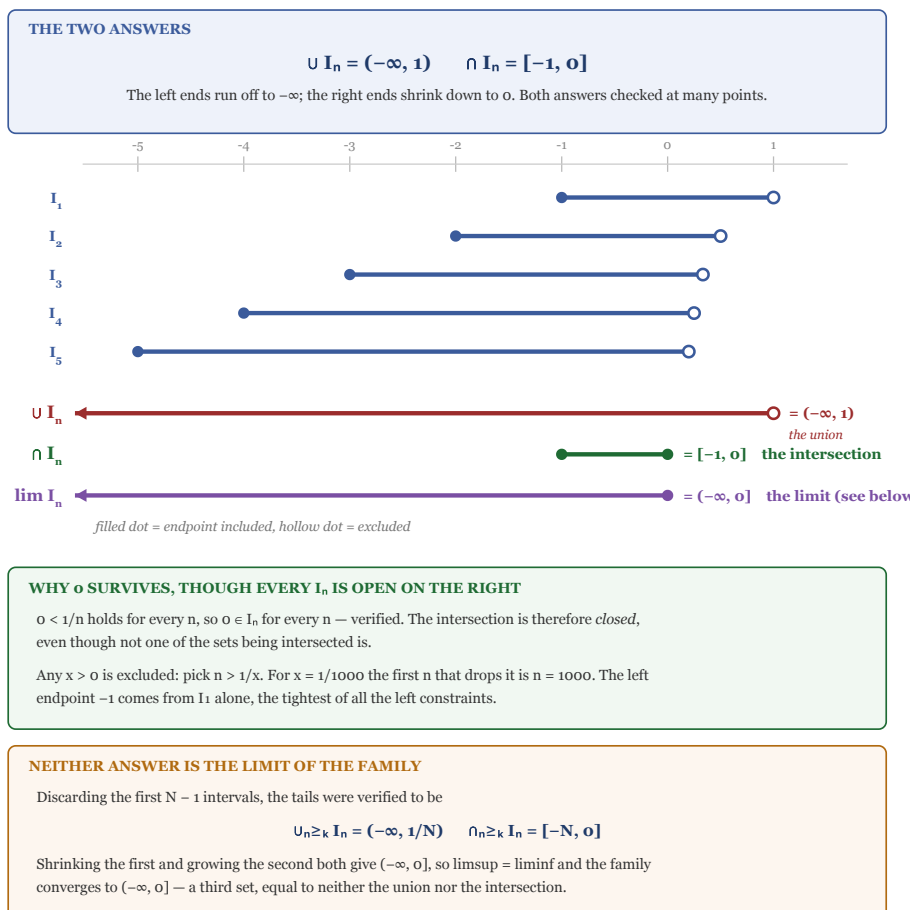


Figure 1.1: A diagram in four parts. The first gives the two answers: the union of the intervals I_n is the open interval from minus infinity to 1, and the intersection is the closed interval from minus 1 to 0. The left ends run off to minus infinity while the right ends shrink down to 0, and both answers were checked at many points. The second part is a number line from minus 5 to 1 with the first five intervals drawn as blue bars, each with a filled dot at its left endpoint minus n and a hollow dot at its right endpoint one over n , so I_1 runs from minus 1 to 1, I_2 from minus 2 to one half, I_3 from minus 3 to one third, I_4 from minus 4 to one quarter and I_5 from minus 5 to one fifth. Below them a red bar with a left arrow and a hollow right dot at 1 marks the union as the interval from minus infinity to 1. Below that a green bar with filled dots at both minus 1 and 0 marks the intersection as the closed interval from minus 1 to 0. Below that a purple bar with a left arrow and a filled dot at 0 marks the limit of the family as the interval from minus infinity to 0 with 0 included. A note explains that a filled dot means the endpoint is included and a hollow dot means it is excluded. A green band explains why 0 survives although every interval is open on the right: zero is less than one over n for every n , so zero belongs to every interval, as verified, and the intersection is therefore closed even though not one of the sets being

intersected is. Any x greater than zero is excluded by picking n greater than one over x , and for x equal to one thousandth the first n that drops it is n equal to one thousand, while the left endpoint minus 1 comes from I one alone, the tightest of all the left constraints. An orange band explains that neither answer is the limit of the family: discarding the first N minus 1 intervals, the tails were verified to be the union from n at least N equals the interval from minus infinity to one over N , and the intersection from n at least N equals the closed interval from minus N to 0. Shrinking the first and growing the second both give the interval from minus infinity to 0 with 0 included, so the limit superior equals the limit inferior and the family converges to that set, a third set equal to neither the union nor the intersection.

1.6 Remark — The Family Is Not Nested, and Both Ends Move

A common expectation is that an indexed family of intervals either grows or shrinks, so that the union is the “largest” member and the intersection the “smallest”. Neither happens here. The two endpoints move in opposite senses:

	left endpoint $-n$	right endpoint $\frac{1}{n}$
as n grows	decreases to $-\infty$	decreases to 0
effect on I_n	gains ground	loses ground

So no two intervals are comparable. Checked directly: $0.7 \in I_1$ but $0.7 \notin I_2$, while $-1.5 \in I_2$ but $-1.5 \notin I_1$, so neither $I_1 \subseteq I_2$ nor $I_2 \subseteq I_1$.

The consequence is that neither answer is a member of the family. The union $(-\infty, 1)$ is unbounded and so is no I_n ; the intersection $[-1, 0]$ is closed and no I_n is. Each answer takes the **best left endpoint from one member and the best right endpoint from another**: the union borrows its right end from I_1 , and the intersection borrows its left end from I_1 as well, while its right end comes from the whole tail at once.

1.7 Remark — How 0 Survives an Intersection of Half-Open Sets

The most instructive point in the problem is that

$$0 \in \bigcap_n I_n$$

even though every I_n excludes its right endpoint. There is no contradiction: 0 is never anybody's right endpoint. The right endpoints are $1, \frac{1}{2}, \frac{1}{3}, \dots$, all strictly positive, so 0 sits comfortably inside each interval — verified for all n up to five thousand, and in general because $\frac{1}{n} > 0$.

What fails is the naive rule “the intersection ends where the endpoints converge, and inherits the open end”. The correct statement is that $\inf_n \frac{1}{n} = 0$ is **not attained**, which is exactly what allows the strict inequality $x < \frac{1}{n}$ to survive at $x = 0$ for every n . Had some interval been $[-n, 0)$, the point 0 would have been lost.

This produces a small topological surprise: a countable intersection of sets, none of which is closed, is a closed set. Dually the union $(-\infty, 1)$ is open even though no I_n is open. Arbitrary unions of open sets are open and arbitrary intersections of closed sets are closed, but the reverse inheritances simply do not hold, and this one family breaks both.

1.8 Remark — The Family Converges, but to Neither Answer

Discarding the first $N - 1$ intervals gives the tails, verified for several N :

$$\bigcup_{n \geq N} I_n = (-\infty, \frac{1}{N}), \quad \bigcap_{n \geq N} I_n = [-N, 0].$$

The first shrinks as N grows and the second grows, and they meet:

$$\begin{aligned} \limsup_n I_n &= \bigcap_N \bigcup_{n \geq N} I_n = \bigcap_N (-\infty, \frac{1}{N}) = (-\infty, 0], \\ \liminf_n I_n &= \bigcup_N \bigcap_{n \geq N} I_n = \bigcup_N [-N, 0] = (-\infty, 0]. \end{aligned}$$

Since the two agree, the set limit exists:

$$\lim_{n \rightarrow \infty} I_n = (-\infty, 0].$$

This is a **third** set, neither the union nor the intersection. Equivalently, the indicator functions converge pointwise, $\mathbf{1}_{I_n} \rightarrow \mathbf{1}_{(-\infty, 0]}$ — spot-checked at $x = -5, -1, 0, \frac{1}{100}, \frac{1}{2}$ for n near **900**, where the first three give **1** and the last two give **0**.

The distinction matters. The union records everything that happens **ever**, the intersection everything that happens **always**, and the limit everything that happens **eventually**. Only the last is a limit in any analytic sense, and only for it is the point 1^- irrelevant: the numbers just below 1 belong to I_1 and to nothing else, so they enter the union but are forgotten by the limit.

1.9 Remark — De Morgan Extends to Infinite Families

The complements behave exactly as the finite laws of 4.3 Problem 3 predict:

$$\overline{\bigcup_n I_n} = \bigcap_n \overline{I_n}, \quad \overline{\bigcap_n I_n} = \bigcup_n \overline{I_n}.$$

Concretely $\overline{\bigcup_n I_n} = [1, \infty)$, confirmed at every test point, and indeed a real number fails to lie in any I_n exactly when it is at least 1 .

The reason the finite proof carries over verbatim is that the element-chasing argument never counted anything. Its middle step was

$$\neg(\exists n : x \in I_n) \iff \forall n : \neg(x \in I_n),$$

the quantifier form of De Morgan's law, which holds for index sets of any size. This is the payoff of having proved 4.3 Problem 3 by equivalences rather than by truth tables: truth tables need finitely many rows, whereas the quantifier argument does not care how many n there are.